

Gudipati Manikesh

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PROFESSIONAL SUMMARY

Software Development Engineer with a strong foundation in Data Structures, Algorithms, Object-Oriented Design, and complexity analysis, demonstrated through 90+ solved DSA problems and a 5-Star HackerRank rating. Experienced in designing and shipping production-grade, cloud-native distributed systems, including a scalable examination platform serving 8,000+ users and an enterprise multi-agent AI platform, using Python, Java, and modern cloud services. Passionate about solving ambiguous, large-scale engineering problems and building fault-tolerant, low-latency systems in fast-paced environments.

EDUCATION

VNR Vignana Jyothi Institute of Engineering and Technology - JNTUH <i>B.Tech in Computer Science and Engineering (AIML) — CGPA: 8.2/10</i>	Hyderabad, India 2021 – 2025
Narayana Junior College - TSBIE <i>Intermediate — Percentage: 97.6%</i>	Hyderabad, India 2019 – 2021
Kendriya Vidyalaya Sangathan - CBSE <i>Schooling — Percentage: 86.4%</i>	Nalgonda, India 2010 – 2019

EXPERIENCE

Software Development Engineer (AI Engineer) <i>Tata Consultancy Services (TCS Prime)</i>	2025 – Present India
- Architected and shipped production distributed systems at scale, including a cloud-native examination platform and an enterprise multi-agent AI platform (detailed under Projects), designing solutions from broadly defined problems to deployment. - Built scalable backend services and REST APIs using Python and FastAPI, deployed across Google Cloud Functions and Azure, applying object-oriented design principles for maintainability and fault tolerance. - Worked in an agile environment with cross-disciplinary teams to deliver high-quality, production-ready software on weekly development cycles. - Built and maintained CI/CD pipelines using GitHub Actions, applying version control and testing best practices across all production releases.	
Research Intern <i>Indian Institute of Technology Guwahati</i>	May 2024 – June 2024 Guwahati, India
- Built an NLP based sentiment analysis system on 9000+ TikTok reviews containing 5040 positive, 2835 neutral, and 1125 negative samples. - Achieved 97% accuracy and 0.97 weighted F1-score using TF-IDF vectorization, SVM classification, preprocessing, and sampling techniques. - Evaluated model performance using classification reports, confusion matrix analysis, AUROC score evaluation, and visualization techniques.	
Data Science Intern <i>Null Class Pvt. Ltd.</i>	April 2024 – May 2024 Remote
- Developed age and gender detection models using datasets containing 12391 male and 11317 female image samples with Python and Scikit-learn. - Achieved 90%+ prediction accuracy through preprocessing, feature engineering, and machine learning based classification workflows. - Built and deployed an interactive Streamlit based UI for real time age and gender prediction and visualization.	

PROJECTS

Maitree Online Examination Platform <i>Production System</i> <i>React.js, FastAPI, REST APIs, Google Cloud Functions, Google Cloud Storage</i>
- Architected and deployed an end-to-end, 100% cloud-native online examination platform enabling 8,000+ employees across multiple countries and time zones to take assessments seamlessly. - Built a scalable React.js frontend and FastAPI backend with REST APIs, deploying services on Google Cloud Functions with data stored in Google Firestore and the frontend hosted on Azure. - Delivered a dual-platform solution (Web + Mobile App) backed by a common API layer, supporting thousands of concurrent users with high availability and low latency during peak traffic.

- Built role-based administrative dashboards providing real-time region-wise, score-wise, and performance analytics, reducing manual result-processing time and improving operational visibility.
- Business Impact: Replaced the legacy center-based examination process, eliminating venue and logistics costs while enabling global participation and increasing engagement across the organization.

AURA AI – Enterprise AI Development Platform | *React.js, FastAPI, LangChain, LangGraph, OpenAI, Google Firestore,*

- Engineered an enterprise-grade AI-powered development platform enabling users to generate websites, perform data analysis, create images, and automate software development workflows from a unified interface.
- Built a scalable React.js frontend and FastAPI backend integrating LangChain, LangGraph, and OpenAI LLMs to orchestrate intelligent multi-agent workflows.
- Implemented role-based AI experiences, delivering personalized capabilities for developers, data analysts, and designers through a single enterprise application.
- Integrated secure file uploads, cloud storage, an in-browser code editor, and deployable AI-generated source code outputs, streamlining end-to-end development workflows.
- Business Impact: Unified multiple AI workflows into a single enterprise platform, reducing dependency on standalone tools and accelerating application prototyping and developer productivity.

TikTok Sentiment Analysis System | *Python, NLP, SVM, TF-IDF, Scikit-learn*

- Built an NLP based sentiment classification system on 9000+ TikTok reviews containing 5040 positive, 2835 neutral, and 1125 negative samples.
- Achieved 97% accuracy and 0.97 weighted F1 score using TF-IDF vectorization, SVM classification, preprocessing, and sampling techniques.
- Evaluated prediction performance using classification reports, confusion matrix visualization, precision recall metrics, and AUROC score analysis.

TECHNICAL SKILLS

Programming Languages: Java, Python, C, C++, SQL, JavaScript, HTML, CSS

Core Computer Science: Data Structures and Algorithms, Object Oriented Design, Complexity Analysis, DBMS, Operating Systems, Computer Networks

Distributed and Cloud Systems: Distributed/Service-Oriented Architecture, REST APIs, Google Cloud Functions, Google Firestore, Azure Cloud, Scalable System Design

AI and Machine Learning: Machine Learning, Deep Learning, NLP, LangChain, LangGraph, Scikit-learn, TensorFlow, Pandas, NumPy

Frameworks and Tools: React.js, FastAPI, Git, GitHub, GitHub Actions, Jupyter Notebook, VS Code

ACHIEVEMENTS

- Secured AIR 5435 in GATE 2025 DA Paper among thousands of candidates across India.
- Received Best Research Paper recognition at ICAMSD 2025 International Conference on AI and ML for Sustainable Development.
- Qualified for Round 2 of Flipkart Grid 5.0, a national level software engineering and innovation competition.
- Attained AIR 5884 in CodeKaze 2023 conducted by Naukri Engineers Week.
- Earned 5-Star Problem Solving badge on HackerRank through consistent algorithmic problem solving performance.
- Participated in multiple national level hackathons including TRI-NIT and NNRG Hackathons focused on AI and software innovation.
- Published and contributed to AI and Machine Learning based academic and research oriented projects.

CODING PROFILES

- **LeetCode** – Solved 90+ DSA problems across Arrays, Trees, Graphs, Dynamic Programming, and Greedy Algorithms.
- **HackerRank** – Earned 5-Star Problem Solving badge through competitive coding and algorithmic challenges.
- **CodeChef** – Participated in competitive programming contests focused on data structures and algorithmic problem solving.
- **Codeforces** – Practicing advanced competitive programming through rated contests and coding challenges.
- **GitHub** – Published AI, NLP, Machine Learning, and scalable software development projects with production focused implementations.
- **LinkedIn** – Engaged with software engineering, AI, and technology communities through professional networking.